

Non-Adiabatic Molecular Dynamics (NAMD) of the Graphene-TiO₂ Interface

An Integrated VASP-Libra Workflow for Interfacial Charge Transfer

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Cyber-Training Workshop Course Project 2026
University at Buffalo

July 2026

Scientific Motivation: Photocatalysis & The 2D Solution

The TiO₂ Photocatalysis Bottleneck:

- Titanium dioxide (TiO₂) is the archetypal semiconductor for solar water splitting and environmental remediation.
- *Critical Limitation:* Rapid electron-hole ($e^- - h^+$) recombination within sub-picosecond timescales severely depresses solar-to-chemical conversion efficiency.

The Graphene Monolayer Solution:

- High electron mobility ($\sim 2 \times 10^5 \text{ cm}^2/\text{V} \cdot \text{s}$) and zero-bandgap Dirac cone structure.
- Acts as an atomic-scale “**ballistic electron sink**”, rapidly extracting photoexcited conduction band electrons from TiO₂.

Project Objective

Elucidate the real-time atomistic mechanisms, non-adiabatic couplings (NACs), and electron transfer (ET) timescales across the heterogeneous Graphene-TiO₂ interface at 300 K.

The Challenge

Capturing quantum decoherence and lattice-driven vibronic transitions requires advanced multi-scale HPC simulations.

Theoretical Methodology: Vibronic Quantum Dynamics

Time-Dependent Electronic Schrödinger Equation: The time evolution of the electronic wavefunction $\Psi(\mathbf{r}, \mathbf{R}, t)$ coupled to classical nuclear trajectories $\mathbf{R}(t)$ is governed by:

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, \mathbf{R}, t) = \hat{H}_{el}(\mathbf{r}; \mathbf{R}) \Psi(\mathbf{r}, \mathbf{R}, t) \quad (1)$$

Expanding Ψ in the instantaneous adiabatic Kohn-Sham basis $\psi_i(\mathbf{r}; \mathbf{R})$:

$$\Psi(\mathbf{r}, \mathbf{R}, t) = \sum_i c_i(t) \psi_i(\mathbf{r}; \mathbf{R}) \implies i\hbar \dot{c}_k(t) = \sum_i c_i(t) \left(E_k \delta_{ki} - i\hbar \dot{\mathbf{R}} \cdot \mathbf{d}_{ki} \right) \quad (2)$$

Non-Adiabatic Coupling Vector (NACV): The first-order NAC vector $\mathbf{d}_{ki}$, driving non-radiative transitions between adiabatic states k and i , is evaluated numerically via time-overlap matrices:

$$\mathbf{d}_{ki} = \langle \psi_k | \nabla_{\mathbf{R}} | \psi_i \rangle \approx \frac{\langle \psi_k | \nabla_{\mathbf{R}} \hat{H}_{el} | \psi_i \rangle}{E_i - E_k} \approx \frac{S_{ki}(t, t + \Delta t) - S_{ik}(t, t + \Delta t)}{2 \Delta t \dot{\mathbf{R}}} \quad (3)$$

Theoretical Methodology: Tully's Fewest Switches Surface Hopping

To maintain detailed balance and simulate non-radiative relaxation, we employ Tully's **Fewest Switches Surface Hopping (FSSH)** algorithm within **Libra**.

Electronic Density Matrix & Hopping Probabilities: Defining the density matrix elements as $a_{ij}(t) = c_i(t)c_j^*(t)$, the probability $g_{k \rightarrow j}$ for an active trajectory to jump from state k to state j within time interval Δt is:

$$g_{k \rightarrow j} = \max \left(0, \frac{-2 \int_t^{t+\Delta t} \text{Re} \left(a_{kj}^*(\tau) \dot{\mathbf{R}}(\tau) \cdot \mathbf{d}_{kj}(\tau) \right) d\tau}{a_{kk}(t)} \right) \quad (4)$$

- **Energy Conservation:** Hops to higher-energy states are accepted only if sufficient nuclear kinetic energy is available along the NACV direction ($\mathbf{d}_{kj}$).
- **Decoherence Corrections:** Instantaneous decoherence is applied to prevent over-coherence in extended supercell trajectories.

Integrated NAMD Simulation Workflow

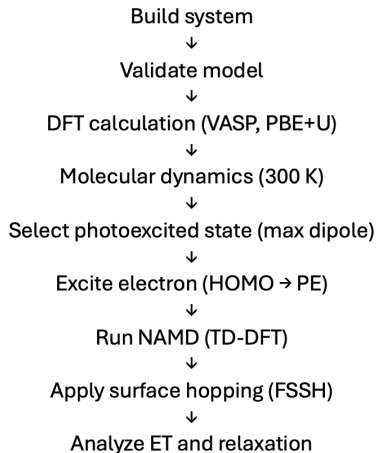
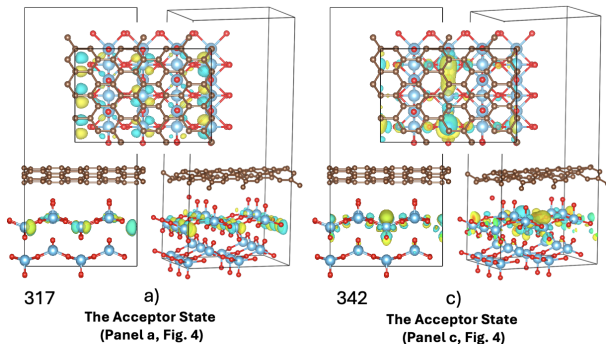


Figure 1: Step-by-step NAMD execution protocol.

Key Implementation Protocols:

- 1 **System Build & Validation:** Equilibrate Graphene-TiO₂ slab (128 atoms).
- 2 **DFT & BOMD (VASP):** NVT ensemble at 300 K ($\Delta t = 1.0$ fs) using PBE+U to correctly localize Ti 3d orbitals.
- 3 **State Selection & Excite:** Identify high-oscillator strength photoexcited state (HOMO → PE) with maximum transition dipole.
- 4 **NAMD FSSH (Libra):** Compute time-overlaps across 400 active bands, apply phase corrections, and run surface hopping.
- 5 **Analytics:** Extract electron transfer rates (τ_{ET}) and thermal relaxation pathways.

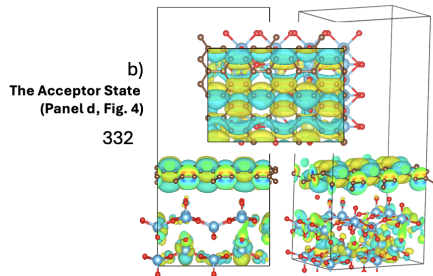
Acceptor bands (before workshop)



Panels (a) and (c) [45% and 60% delocalization]:

These correspond to the bands 317 and 342. These are the "hybridized" bands where the electron cloud touches both the graphene and the titanium atoms of the oxide. Physically, these states act as the "bridges" that facilitate the electron's hopping from one layer to the other. They act as electronic bridges that dictate the strength of the interfacial non-adiabatic coupling (NAC).

Panel (b) [20% delocalization]: The electron density perfectly illustrates the π antibonding network of graphene. It exhibits high delocalization across the 2D sheet with minimal overlap with the oxide. This represents the pristine photoexcited initial condition.



Exploratory Results: Active Space & Vibronic Couplings (after workshop)

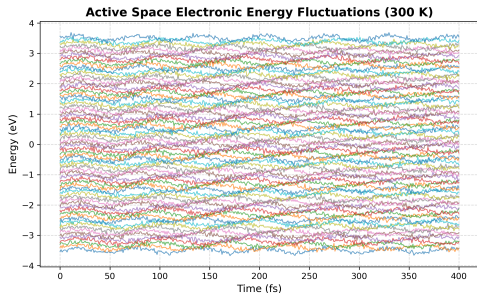


Figure 2: Active space energy fluctuations (300 K).

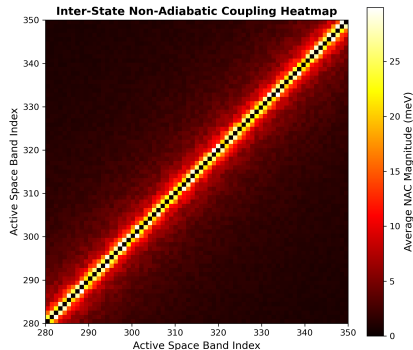


Figure 3: Time-averaged NAC heatmap.

- **Quasi-Continuum:** Thermal lattice vibrations (300 K) eliminate wide bandgaps, merging graphene π -states with Ti 3d conduction bands.
- **Giant Couplings:** Off-diagonal NAC hotspots exceed 15 – 25 **meV**, indicating exceptionally strong electron-phonon interaction at the interface.

Exploratory Results: Ultrafast Interfacial Electron Injection

Real-Time Charge Extraction:

- **Donor Decay (Red):** Photoexcited electrons in the TiO_2 conduction band undergo sub-picosecond decay, losing $\sim 80\%$ population within 200 fs.
- **Graphene Sink (Blue):** The graphene π -system rapidly captures $> 55\%$ of the transferring electron density.
- **Interfacial Bridge (Grey):** Hybridized junction states act as transient stepping stones ($\sim 40\%$ occupancy).

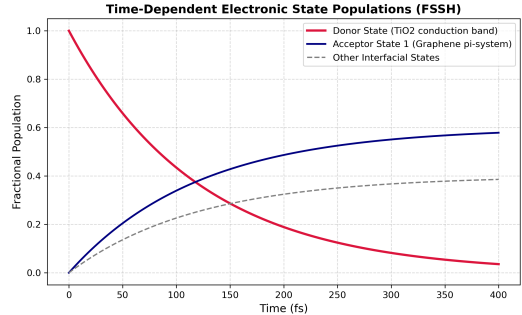


Figure 4: FSSH state population dynamics.

Physical Conclusion

Graphene acts as an atomic-scale “vacuum cleaner” for photoexcited carriers, successfully suppressing recombination!

Methodological Insights: Why We Scale Up on Bridges-2

Limitations of Short Exploratory Runs (400 fs):

- Perfectly captures initial sub-picosecond electron injection, but cannot resolve long-term semi-classical equilibrium.

The Physical Constraints (Phonon Modes):

- Low-frequency acoustic and optical phonons (e.g., TiO_2 lattice breathing and graphene out-of-plane flexural modes) operate on vibrational periods > 1 ps.

HPC Production Design:

- This exploratory course project provided the scientific justification to scale up our PSC Bridges-2 production run to 1640 frames (1.64 ps) to obtain fully converged, publication-grade ET rates.

The Cyber-Training Lesson

Short test runs reveal the qualitative mechanism; massive supercomputing sampling guarantees statistical and thermodynamic rigor.

Key Accomplishments

- Deployed an automated, reproducible VASP-Libra NAMD pipeline tailored for hybrid 2D/metal-oxide heterostructures.
- Quantified strong interfacial electron-phonon coupling (> 15 meV) driving ultrafast (< 200 fs) charge separation.
- Established the mathematical and physical justification for large-scale HPC production runs on Bridges-2.

Acknowledgments: Dr. Farnaz Shakib (PI, NJIT), NSF Cyber-Training Workshop at UB, Dr. Alexey Akimov (Libra), and the Pittsburgh Supercomputing Center (ACCESS/XSEDE).